

From Hubs to Hypervisors: Network Architecture

Networking Concepts · Data Planes · Hypervisors · VLANs

There are moments in learning when everything clicks — and then there are moments when your brain just **melts**, in the best possible way.

This week was one of those moments. I went deep on three interconnected networking concepts that completely reshaped how I think about infrastructure and security:

- → The three operational planes inside every network device
- → How the control plane made entire categories of dedicated hardware obsolete
- → Why a hypervisor is literally a software-defined network running inside your computer
- → Why confusing VLANs with subnets can open serious security gaps

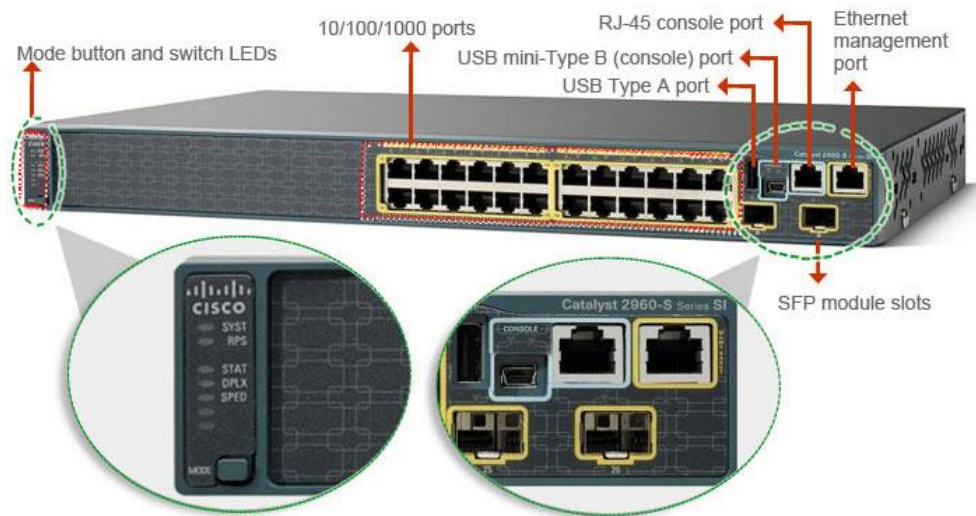
Let me break it all down.

Part 1: The Three Planes of Every Network Device

Every modern network device — a switch, a router, a firewall — is structured around three distinct planes. Think of them as layers of intelligence stacked inside the same box:

- Data Plane (Forwarding Plane) — This is where packets actually move. Wire-speed forwarding, NAT, VLAN tagging, QoS. Hardware-level operation for maximum throughput.
- Control Plane — The "brain." Maintains routing tables, ARP cache, and MAC address tables. Makes all the forwarding decisions that the Data Plane then executes.
- Application Plane (Management Plane) — Where administrators define policies through CLI, SSH, SNMP, or REST APIs. These configurations are pushed down to the Control Plane. (**Important:** do not confuse this with OSI Layer 7, which is also called the "Application Layer" — completely different concept, different context.)

Many enterprise devices expose dedicated physical management ports to keep administrative control traffic fully isolated from user data on the wire. Even the traffic between planes can be separated at the hardware level.



Source of the image: <https://blog.router-switch.com/2017/07/console-port-vs-management-port-in-networking-devicescisco-2960s/>

Part 2: From Hubs to Smart Switches — The Evolution That Changed Everything

Old-school hubs were brutally simple: packet arrives → hub broadcasts it to every single node on the network. No intelligence. No filtering. Just noise — and a serious security problem, since every device on the segment could see every packet.

Modern **switches** changed everything. They maintain **MAC address tables** — mapping each device's hardware address to the specific physical port it's connected to. When a packet arrives, the switch looks up the destination MAC and forwards it only to the correct port. Only the intended device receives the packet.

Clarification: NAT (Network Address Translation — mapping public IPs to private ones) is a Layer 3 function performed by routers, not switches. Basic switches operate at Layer 2 using MAC address tables and do not process IP addresses for NAT purposes.

But here's what genuinely blew my mind: thanks to the control plane, a single properly configured Layer 3 switch today can replace what used to require multiple dedicated hardware appliances — separate firewall boxes, routers, and more. The cost savings in enterprise environments are enormous.

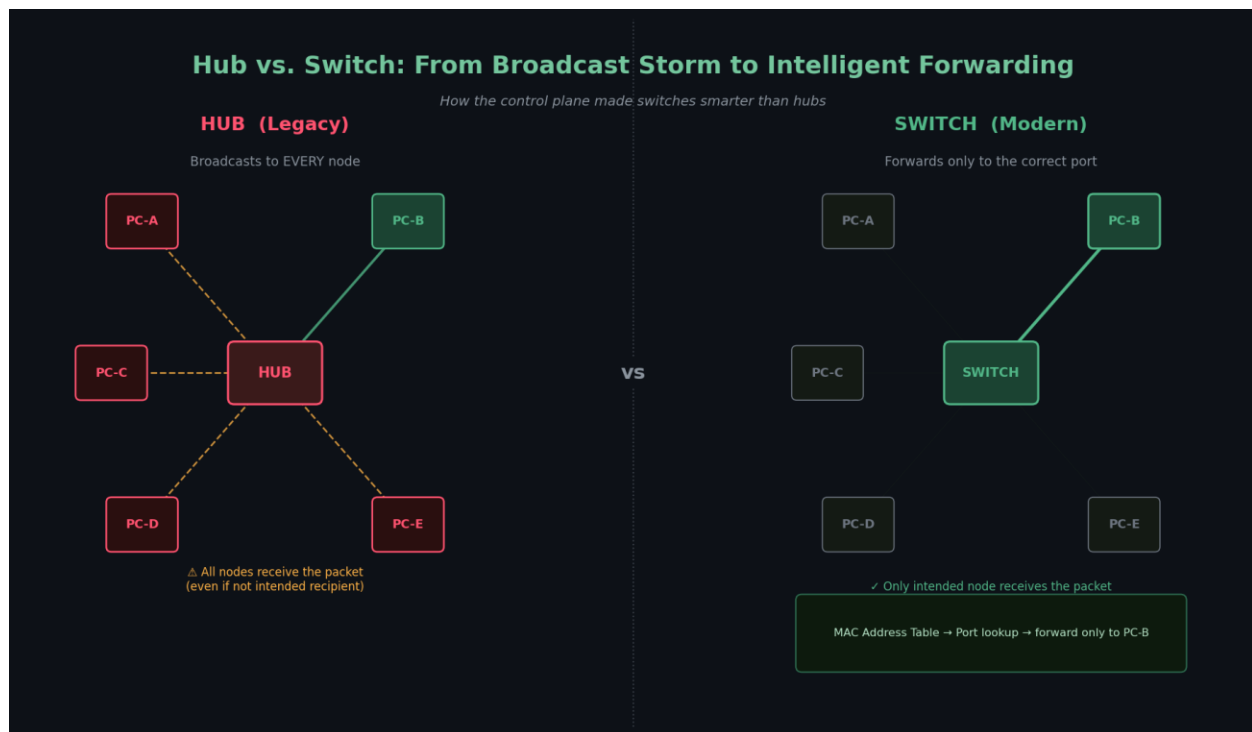


Figure 2 — Hub (broadcast to all) vs. Switch (targeted forwarding via MAC address table)

Part 3: Hypervisors Are Literally a Network Inside a Network

Set a VM's network adapter to "internal network only" in VirtualBox. Simple setting, right? But look at what is actually happening under the hood:

The hypervisor becomes the **control plane** for your VMs. It maintains its own routing tables in RAM, enforces firewall rules, handles NAT, and manages VLAN tagging — all using the physical hardware inside your machine. This is **Network Function Virtualization (NFV)**.

A network running inside another network. On the same hardware. Software literally becomes the switch.

Docker Engine works on a similar principle — but with an important distinction: a VM is a full additional operating system consuming host CPU and RAM, while a container is an isolated application environment that includes only its required dependencies. Same isolation concept, fundamentally different architecture.

Part 4: VLANs vs. Subnets — The Security Difference That Actually Matters

Both VLANs and subnets provide host isolation — but they operate at different OSI layers, and that difference has major security implications:

- Subnets → Layer 3 (Network layer) — IP-based segmentation enforced by routers

- VLANs → Layer 2 (Data Link layer) — Port/tag-based segmentation enforced by switches

Scenario A: Same VLAN, Different Subnets — Security Risk

Attacker and target share the same VLAN but sit in different subnets. The attacker manually changes their IP address to match the target's subnet. This triggers a Layer 2 broadcast — the switch allows it within the VLAN, the target responds, and direct communication is established. The traffic never reaches the router or firewall, because Layer 2 communication does not require routing. The attacker bypasses all Layer 3 security controls entirely.

Scenario B: Different VLANs, Same Subnet — Misconfiguration

This is a misconfiguration. The switch blocks inter-VLAN traffic regardless of the IP address, making communication impossible. To restore connectivity, you would need to either correct the subnet configuration or explicitly route traffic through the router. Not a security scenario — just broken.

Best Practice: Different VLAN + Different Subnet — Fully Secure

In this setup, the **router is the gatekeeper**. Its sub-interfaces are mapped to specific VLANs via trunk ports, and it knows the IP ranges of all subnets. All traffic between hosts must pass through the router — where firewall rules, ACLs, and security policies can be enforced. Full control. Full visibility.

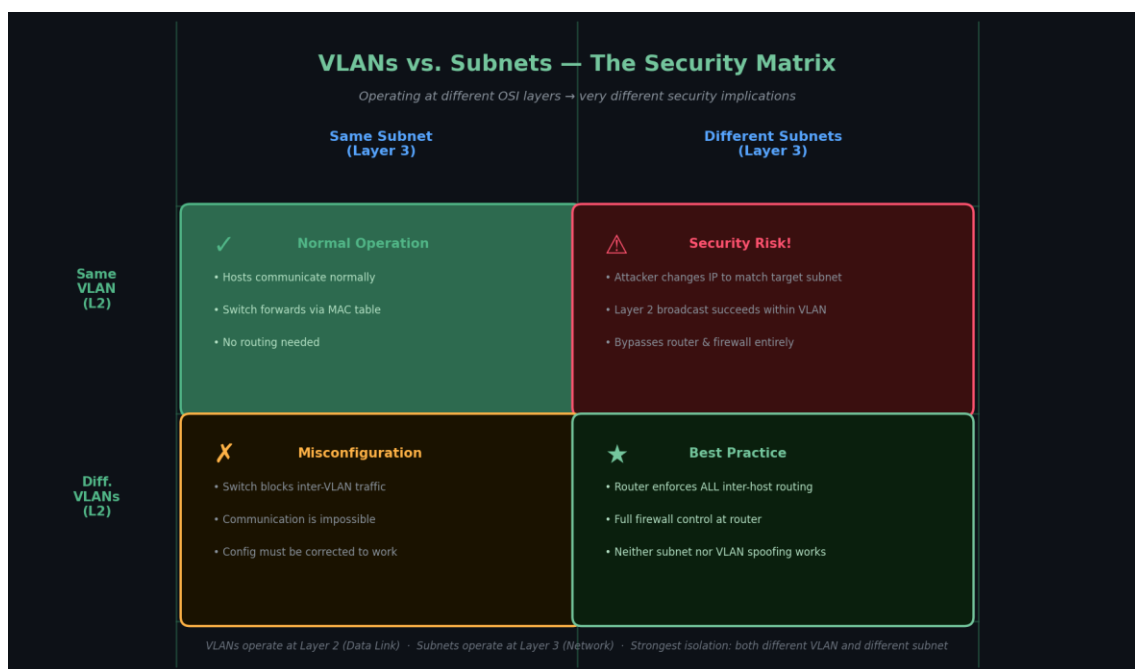


Figure 4 — VLAN/Subnet Security Matrix: four scenarios and their security implications

Key Takeaway

Networking is not just about moving packets. It's about understanding **where** decisions are made (data vs. control vs. management plane), **who** makes them (switches vs. routers vs. hypervisors), and **how** isolation is actually enforced (VLANs vs. subnets).

These concepts appear constantly in cybersecurity — cloud misconfigurations, penetration testing, threat modeling, and network segmentation design. Understanding them deeply is what separates people who configure networks from people who can break, defend, and architect them.